Is protein structure a molecular clock?

Revisiting structural divergence with AlphaFold, orthology and divergence times

An AI-led Phase 2 project of the Beltrao-lab agentic AI hackathon · [presenter] · [date]

Disclosure: every result here was produced by AI

!

Exercise in assessing the bioinformatics capabilities of agentic AI (Beltrao-lab hackathon, github.com/evocellnet/ai-hackathon).

Executed by Claude Opus 5.5 with Opus/Sonnet 5.5 subagents, within a 48 h compute cap (48 CPUs, 500 GB RAM, 1 A100).

Results have not yet been verified by a human. This is not a traditional publication.

Repository: [link to project repository] · Report: technical report/report.md

1

The hackathon

Can an AI agent run a bioinformatics project from prompt to paper?

A week-long test of agentic AI in bioinformatics

Beltrao lab

ETH Zürich

Colm Ryan lab

UCD

Emanuel Gonçalves lab

IST

Christophe Dessimoz lab

UNIL

2 phases

Phase 1: 'predict X' tools
Phase 2: open discovery

15

Phase-1 tools listed (pockets, docking, orthology, kinases, MS…)

Claude

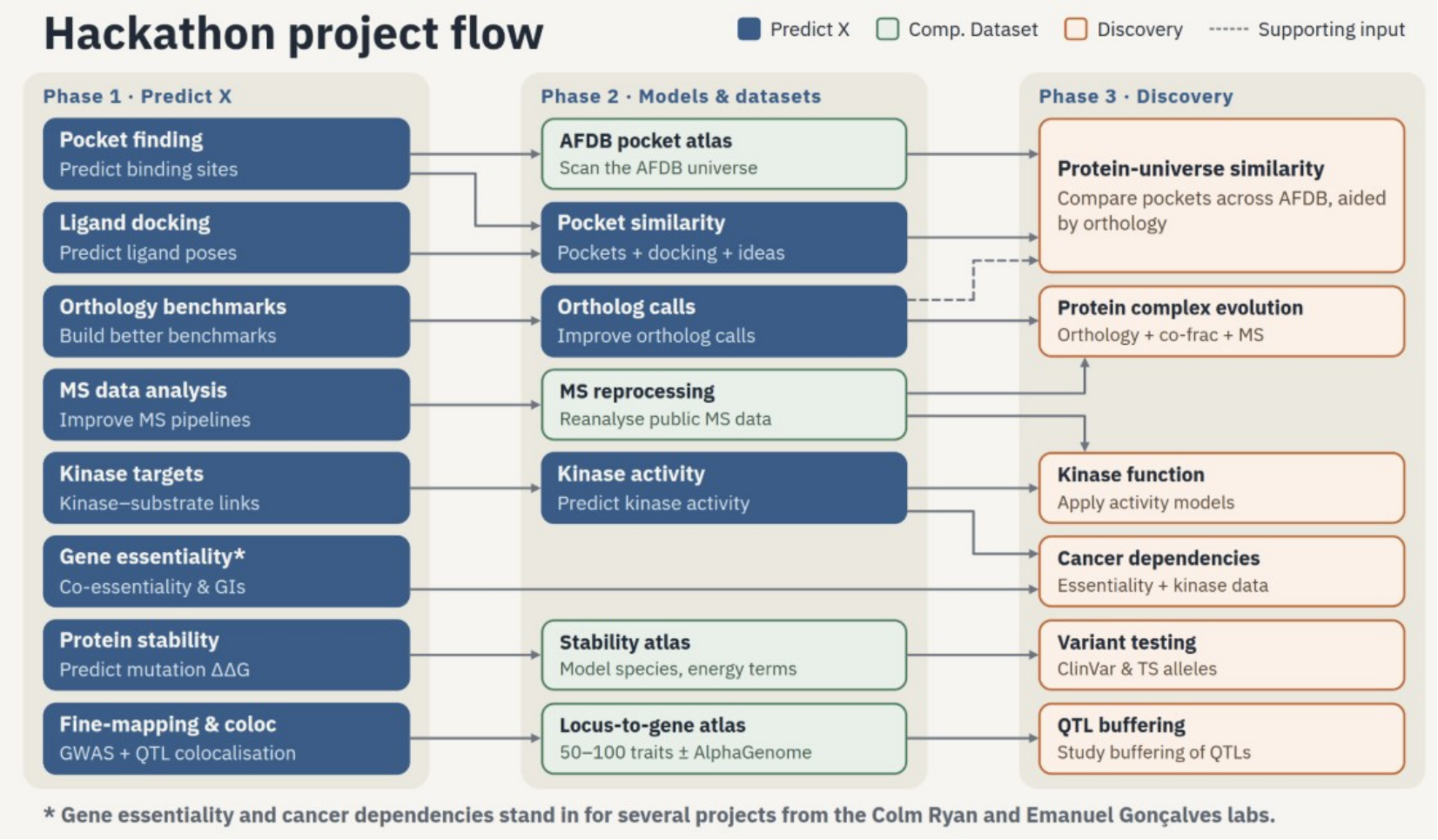
Opus 5.5 as PI agent, delegating to Opus/Sonnet subagents

[__]

participants · [dates] · [Euler cluster details]

Humans verify methodology and benchmarking afterwards; each repository opens with a human-written disclosure.

From prediction tools to discovery projects



This project

Phase 2 discovery: does protein 3D structure evolve like a clock?

Built on

Ancestra (Phase-1 orthology tool) + AlphaFold DB clusters (Foldseek) + TimeTree

2

The prompt

One generic research scaffold + one project-specific question

The prompt has two layers

Shared by all projects

PI role

plan, delegate, review, decide

Design by critique

literature → design → ≤3 review rounds

Reproducibility

seeds, DATA.md, git, no hard-coding

Memory

research tree + lessons file

Verification

toy data, controls, non-author review

Report

paper-style, disclaimer, figures

Specific to this project

Revisit Elofsson 2009

linear, 3–10× more conserved?

AFDB clusters 2023

evolution never analysed

Use Ancestra

orthologs vs paralogs

Neutral clock?

comment on structure

Ideas (priors)

LDO, kinases, cytoskeleton

Budget

48 h, node move at 12 h

Two goals, five questions

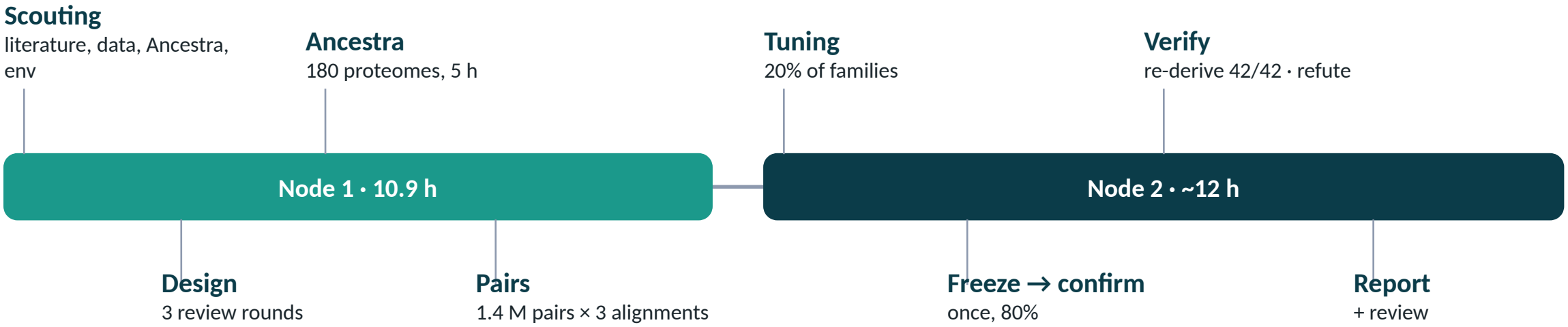
Goal 1 · Is structure a molecular clock?

1. Shape: linear or saturating vs sequence?
2. Ratio: how much slower than sequence?
3. Time: same ratio on young and old branches?
4. Lineages: structure-specific rates?

Goal 2 · AFDB clusters in the light of evolution

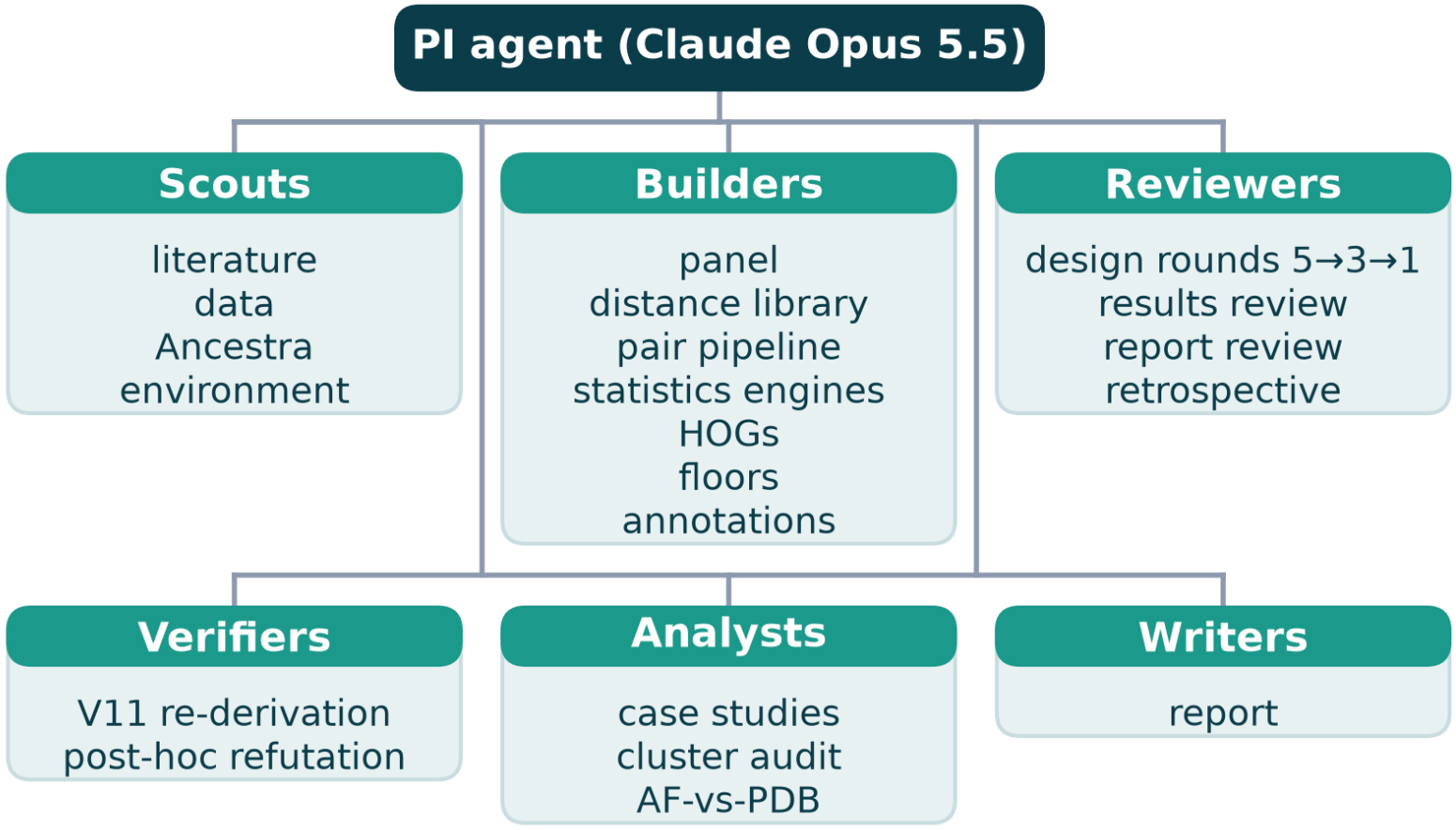
5. Duplication: do paralogs diverge more in structure?
6. Do structural clusters match ortholog groups?
7. Case studies: cytoskeleton & kinases

24 hours of wall clock, two compute nodes



Node move at 20:30 on day 1 (checkpointed); resumed from the research tree on day 2.

An AI research team: one PI agent, many specialists



≈40

subagent tasks with written reports

13

independent reviews (design, results, report, retrospective)

85+

git commits: every decision traceable

3

How the method was decided

Scout → design → independent critique → pilot → freeze

Scouting first: four agents in parallel

Literature 1

Elofsson 2009 used substitutions, not time → a time test is new

Data 2

Cluster pairs are pre-filtered by structure → biased toward plateau

Orthology 3

Ancestra: 148/149 Ensembl orthologs; duplication timing only 36% right

Compute 4

Foldseek GPU works; dated trees reliable for eukaryotes only

→ Design choices: eukaryote panel with dated tree · orthologs chosen by sequence only · never trust duplication ages · measure the AlphaFold bias

Design by independent critique: 5 → 3 → 1 reviewers

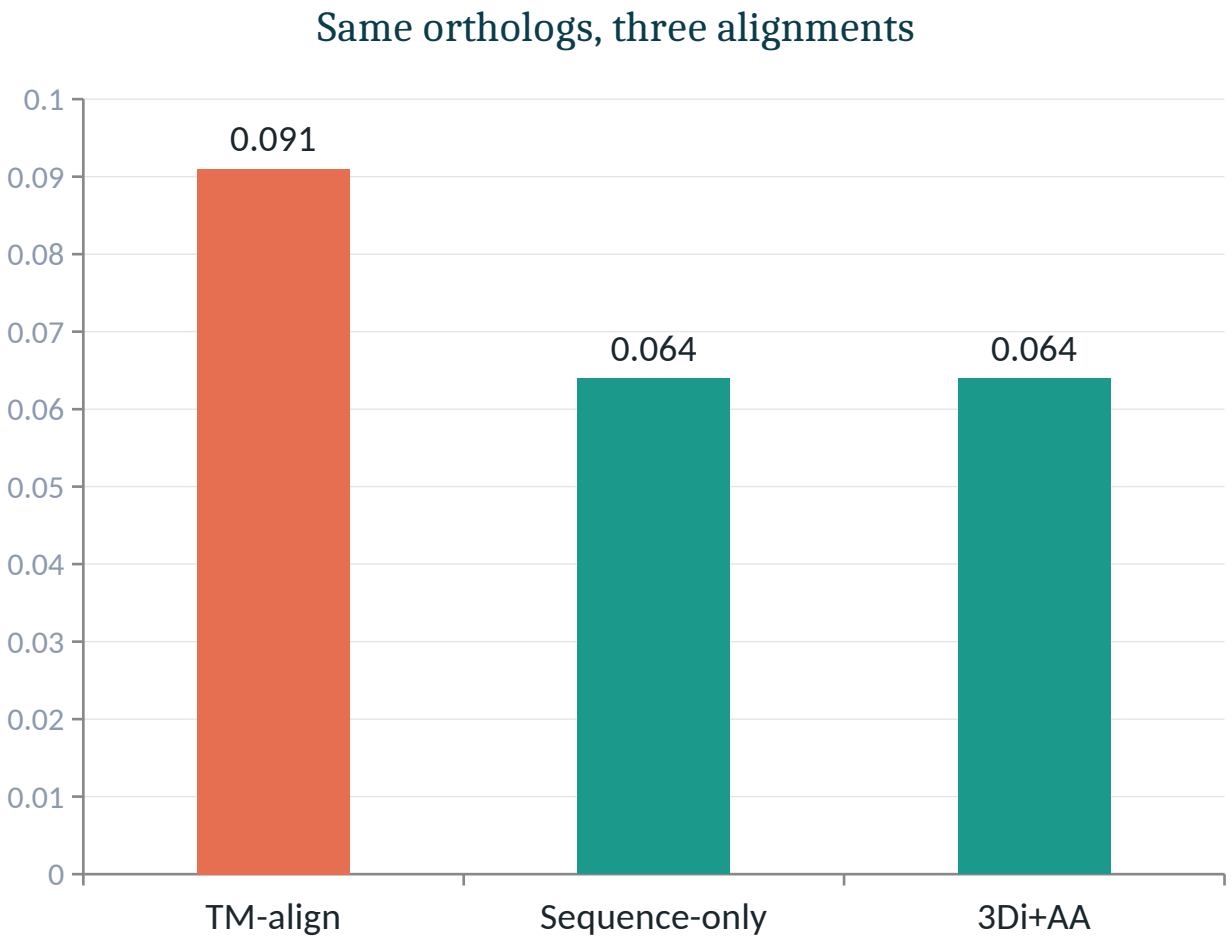


Round 1 **Round 2** **Round 3**

Every reviewer had to propose concrete changes within scope, or say none were needed.

- ✓ Branch lengths on the dated species tree replace pair medians
- ✓ Pre-set decision rules (equivalence margins) replace AIC/BIC
- ✓ Positive control (duplication) + power checks
- ✓ Tune on 20%, confirm once on 80%; Holm correction
- ✓ Noise floors; errors-in-variables correction

Pilots on real data caught what reviewers missed



8 / 104

identical-sequence pairs got non-zero AA distance under TM-align

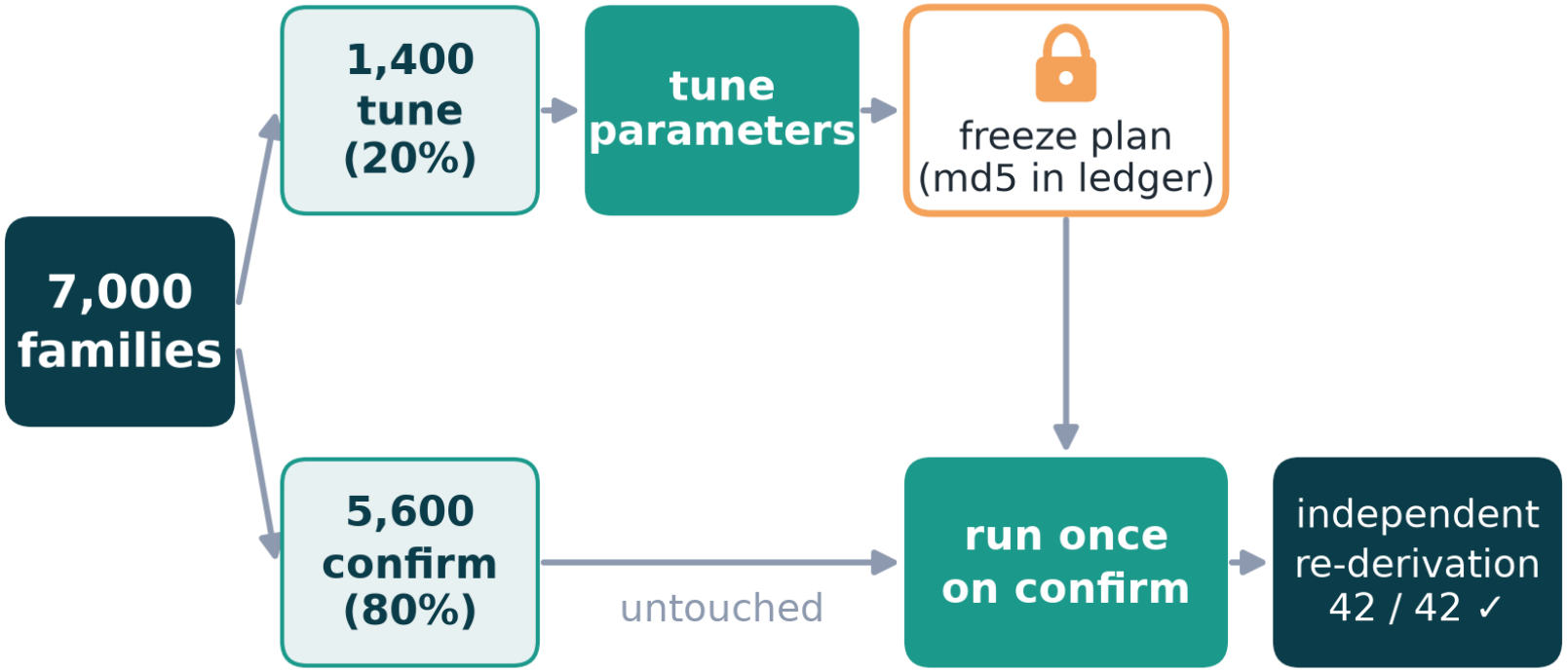
Decision D22

primary alignment → Foldseek 3Di+AA; TM-align and sequence-only kept as brackets

Lesson

no reviewer predicted it; a small real pilot did

Pre-registration: tune on 20%, confirm once on 80%



10/10

automated guards passed before the single confirmation run

34

logged deviations (32 before confirmation)

42/42

numbers re-derived by a non-author agent

Key decisions and why

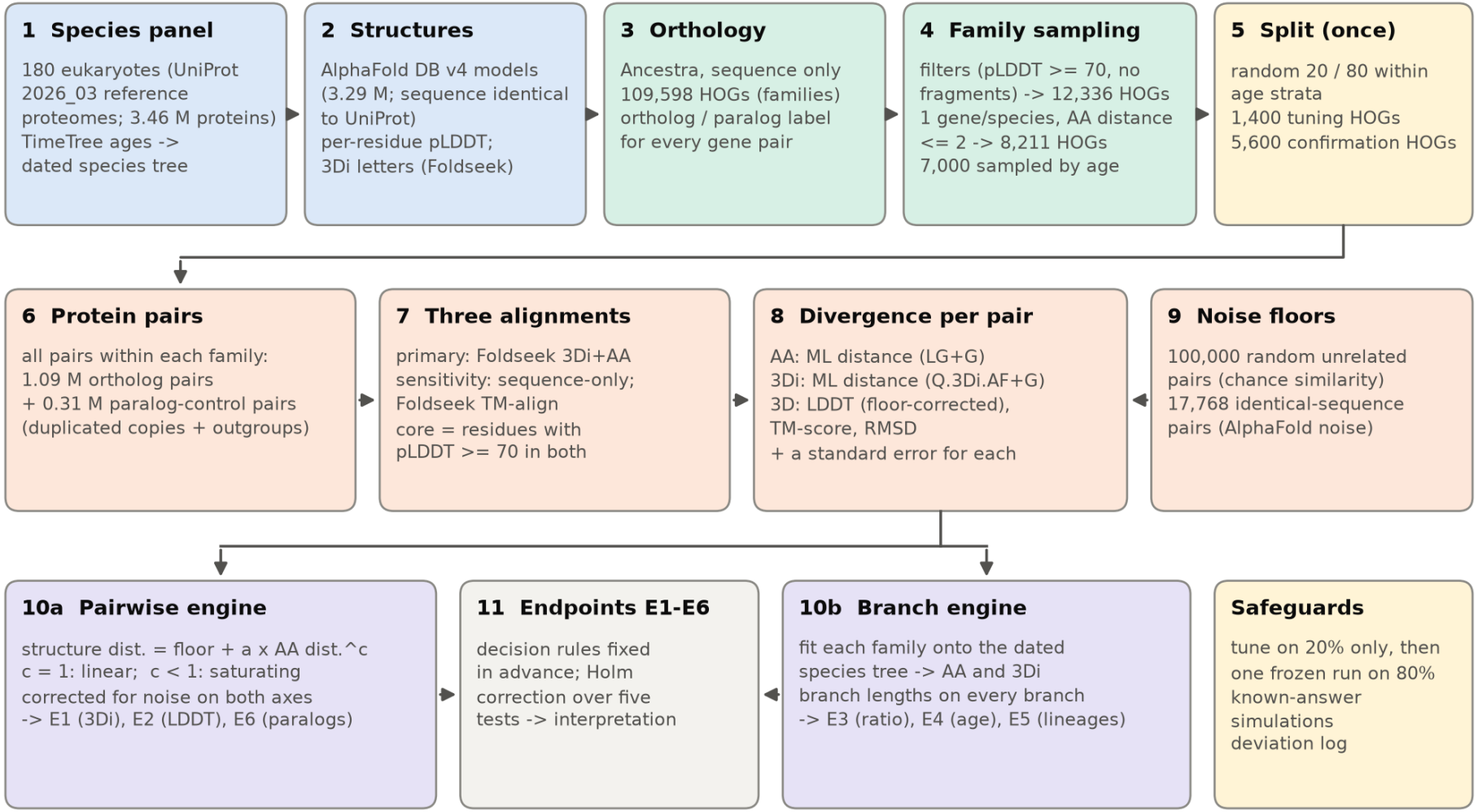
Decision	Why
Use only proteins whose AlphaFold model sequence = UniProt sequence	sequence and structure must describe the same molecule
Never use Ancestra's duplication ages	only 36% agreement with Ensembl
One Ancestra run on 180 whole proteomes	per-cluster runs would cost thousands of CPU-hours
3Di+AA alignment as primary (D22)	TM-align failed the identical-sequence check
Fix two estimator biases found in simulation (D31, D32)	both would have produced false positives
Keep pre-set margins even when power was low	never change tests to make results look better
Report post-hoc reinterpretation separately, after independent refutation attempt	a fixed noise floor turned out incomplete

4

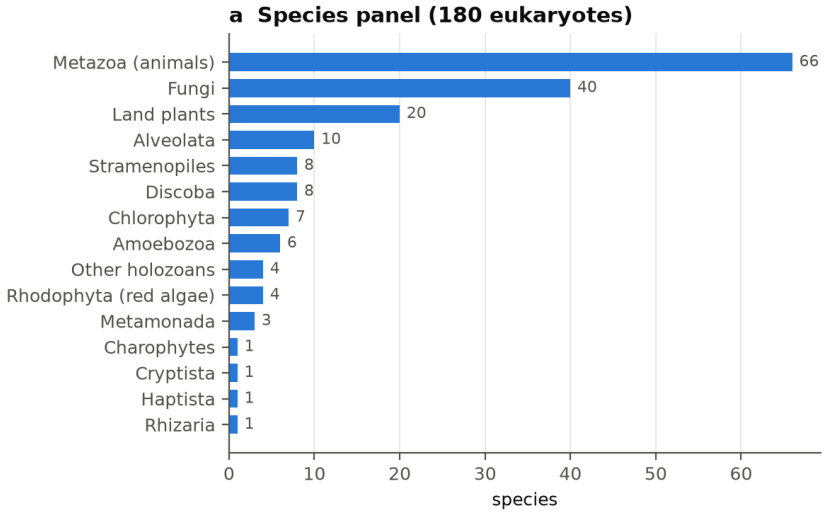
The method, step by step

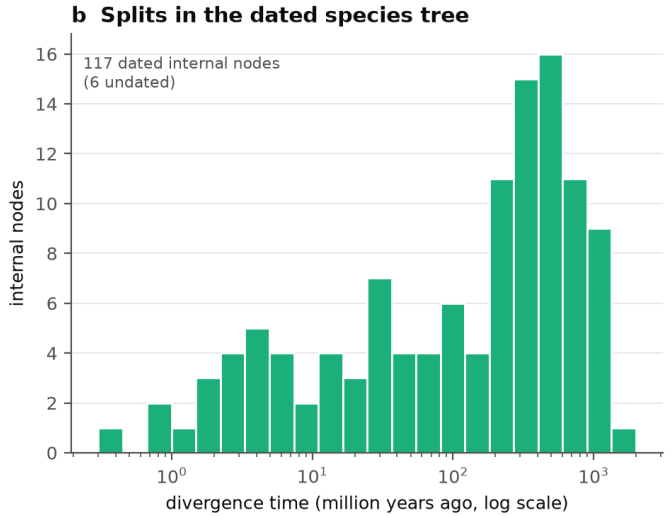
With plain-language explainers for the tricky parts

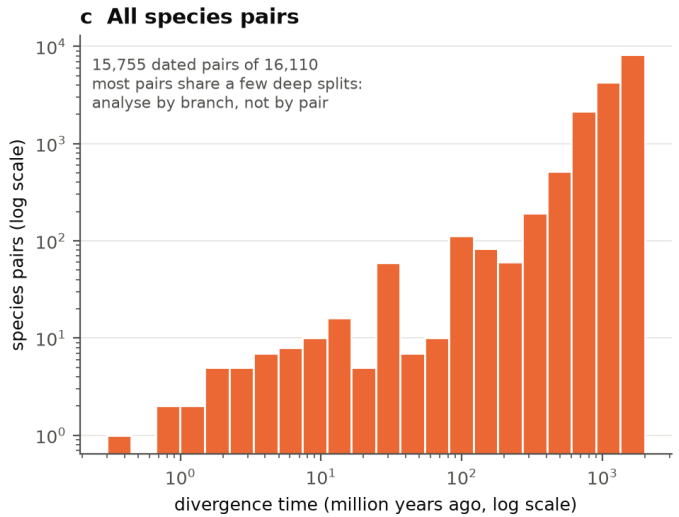
Pipeline at a glance

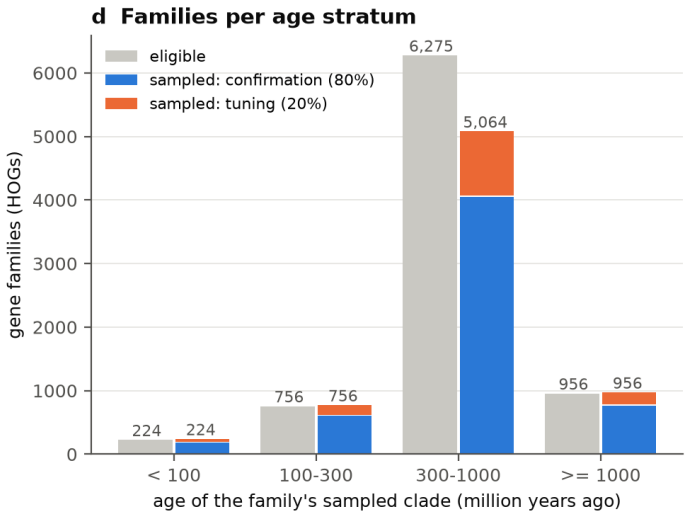


Data: 180 eukaryotes, 3.3 M predicted structures









180

eukaryotic proteomes (66 animals, 40 fungi, 20 plants, 54 protists…)

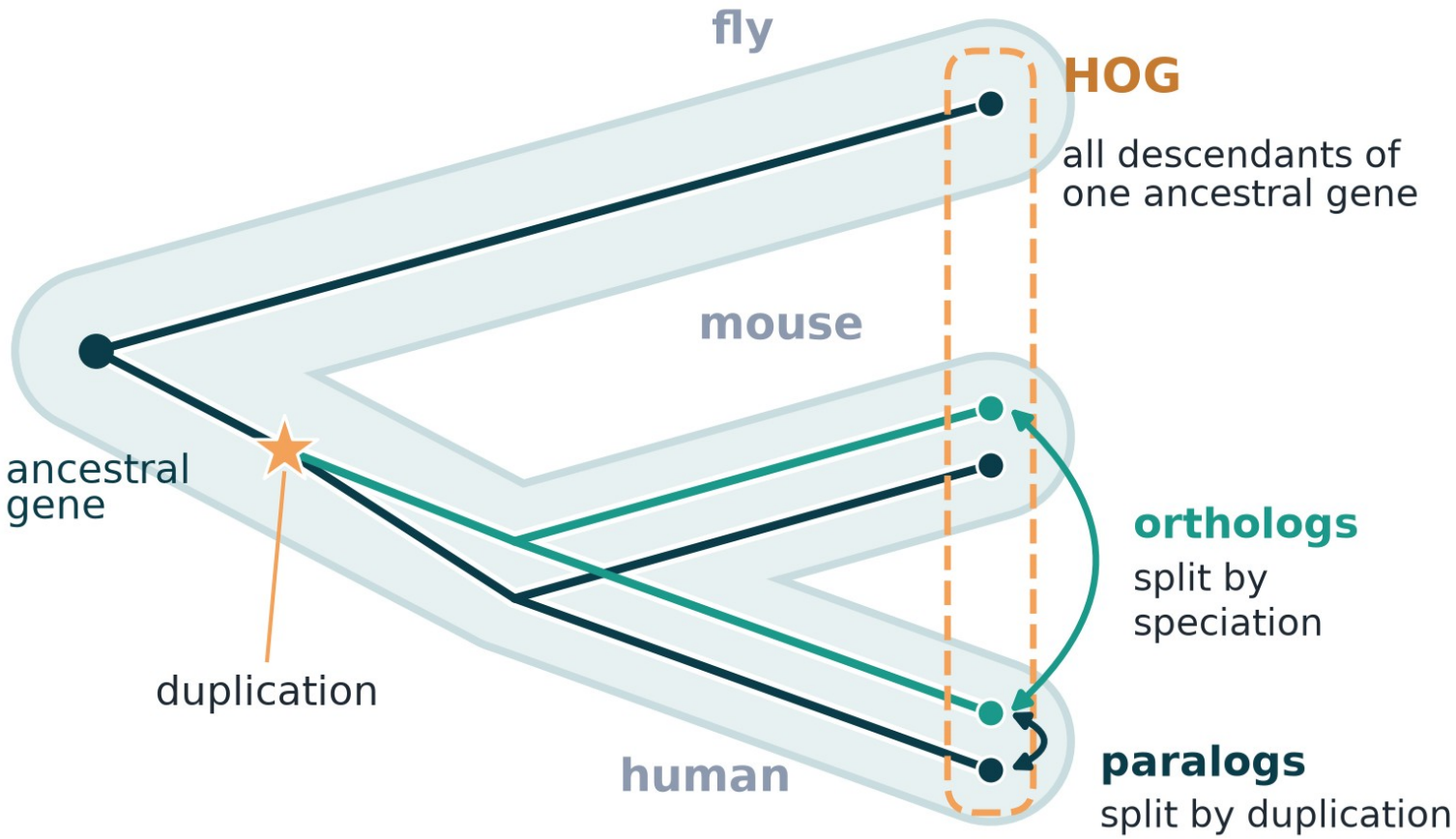
109,598

ortholog families (HOGs) from Ancestra

1.4 M

protein pairs from 7,000 sampled families

Explainer: orthologs, paralogs and HOGs



Ortholog

split by speciation → age known from the species tree

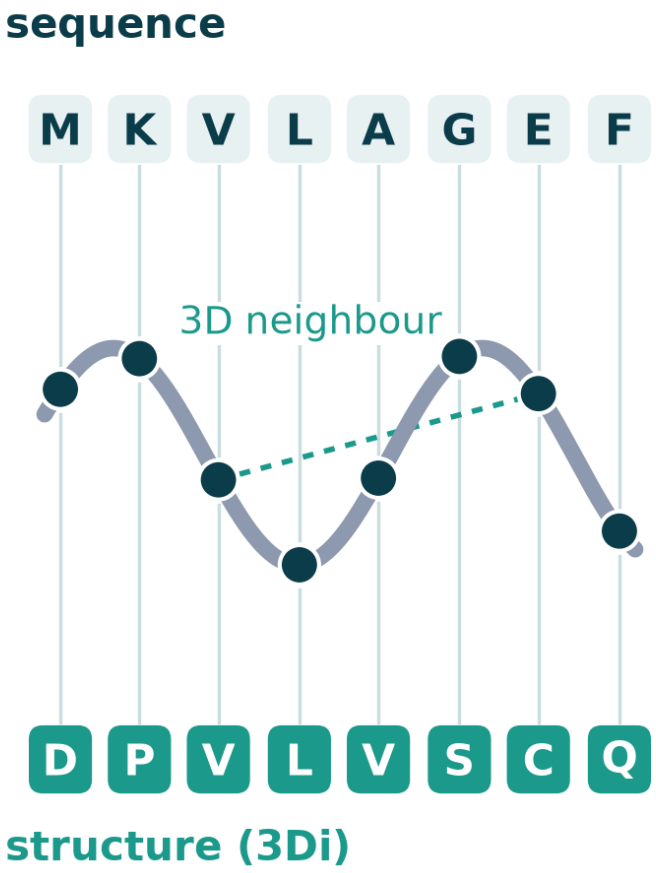
Paralog

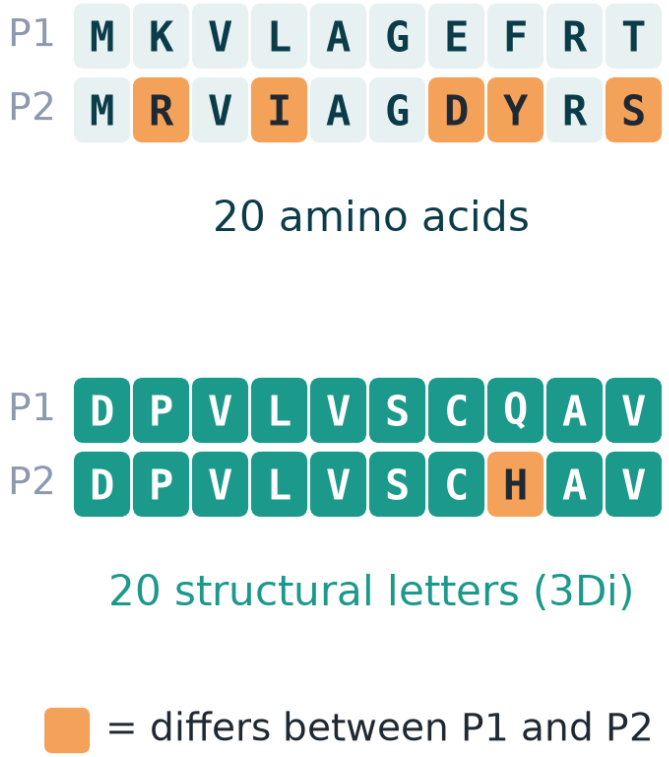
split by duplication → age unknown

Why it matters

orthologs give a clean clock test

Explainer: AlphaFold models and 3Di letters





20 vs 20

amino acids vs structural letters: a like-for-like comparison

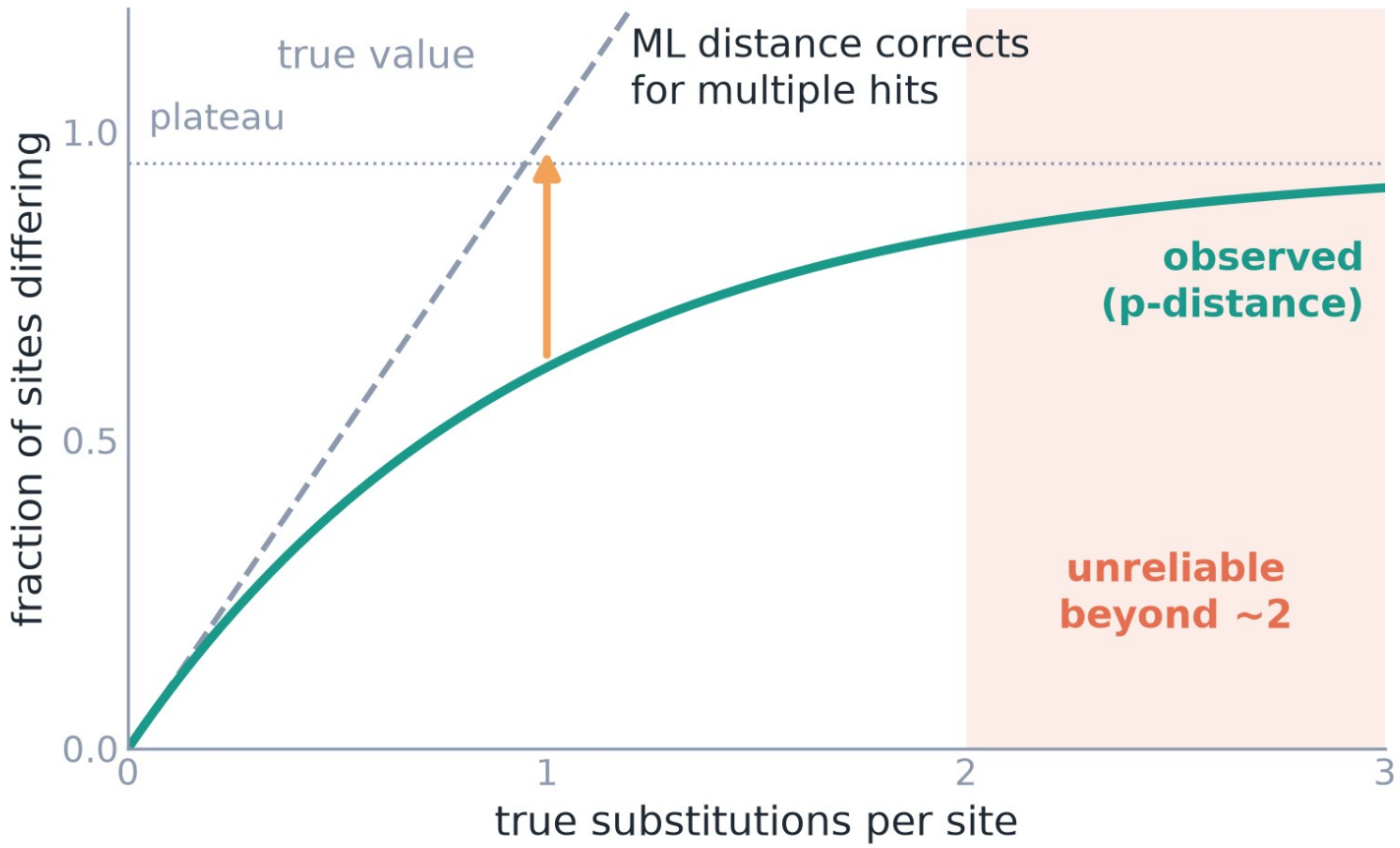
pLDDT ≥ 70

only confidently predicted residues used

Illergård 2009

had to approximate this by grouping amino acids

Explainer: counting change when changes overlap



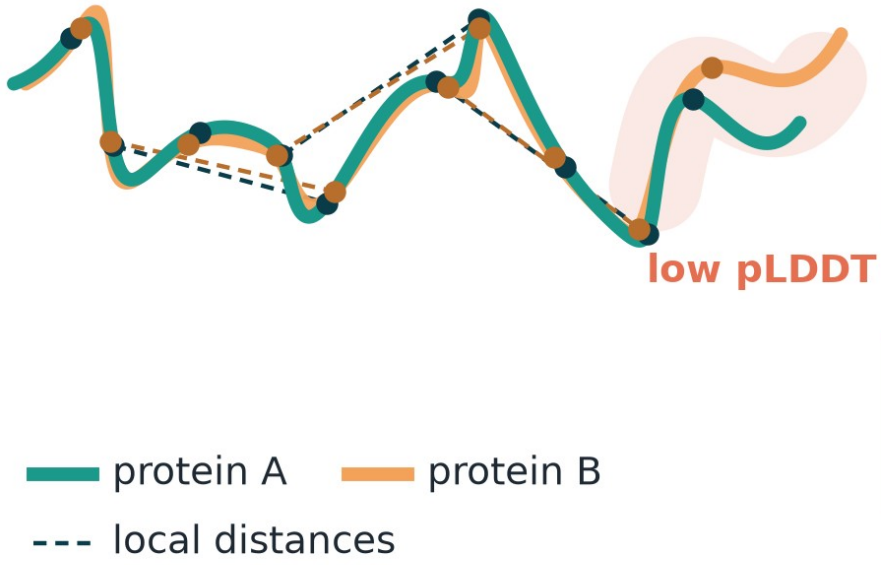
ML distance

expected substitutions per site, corrected for multiple hits

Validated

unbiased within 1.6% on simulated data; matches IQ-TREE within 0.004%

Explainer: three ways to compare two structures



TM-score
global fold match (0–1)

LDDT
fraction of local distances preserved (0–1)

pLDDT
AlphaFold confidence per residue

0.95

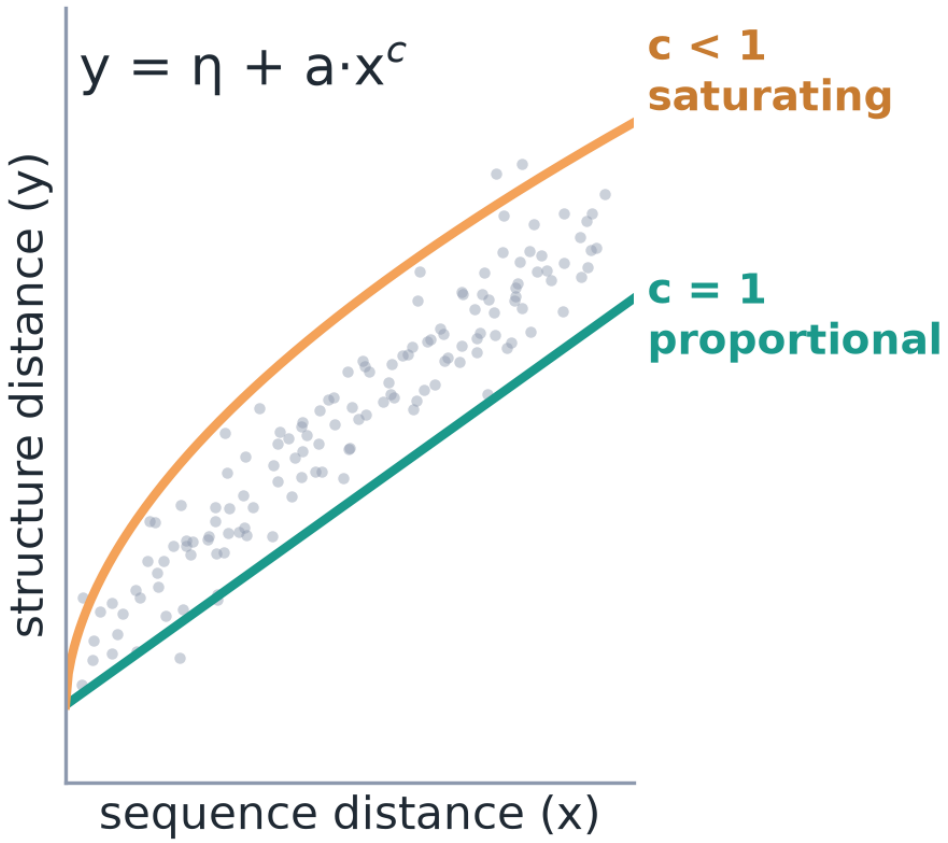
median TM-score between AlphaFold models of identical sequences: predictions are not identical

0.052

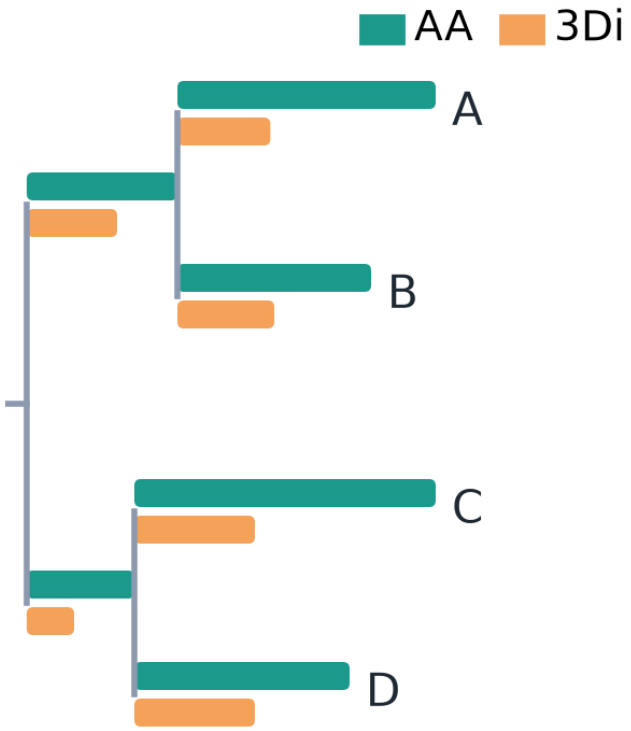
3Di distance at zero sequence change (prediction noise floor)

Explainer: two estimation engines

(a) pairwise engine



(b) branch engine



k = ΣAA / Σ3Di

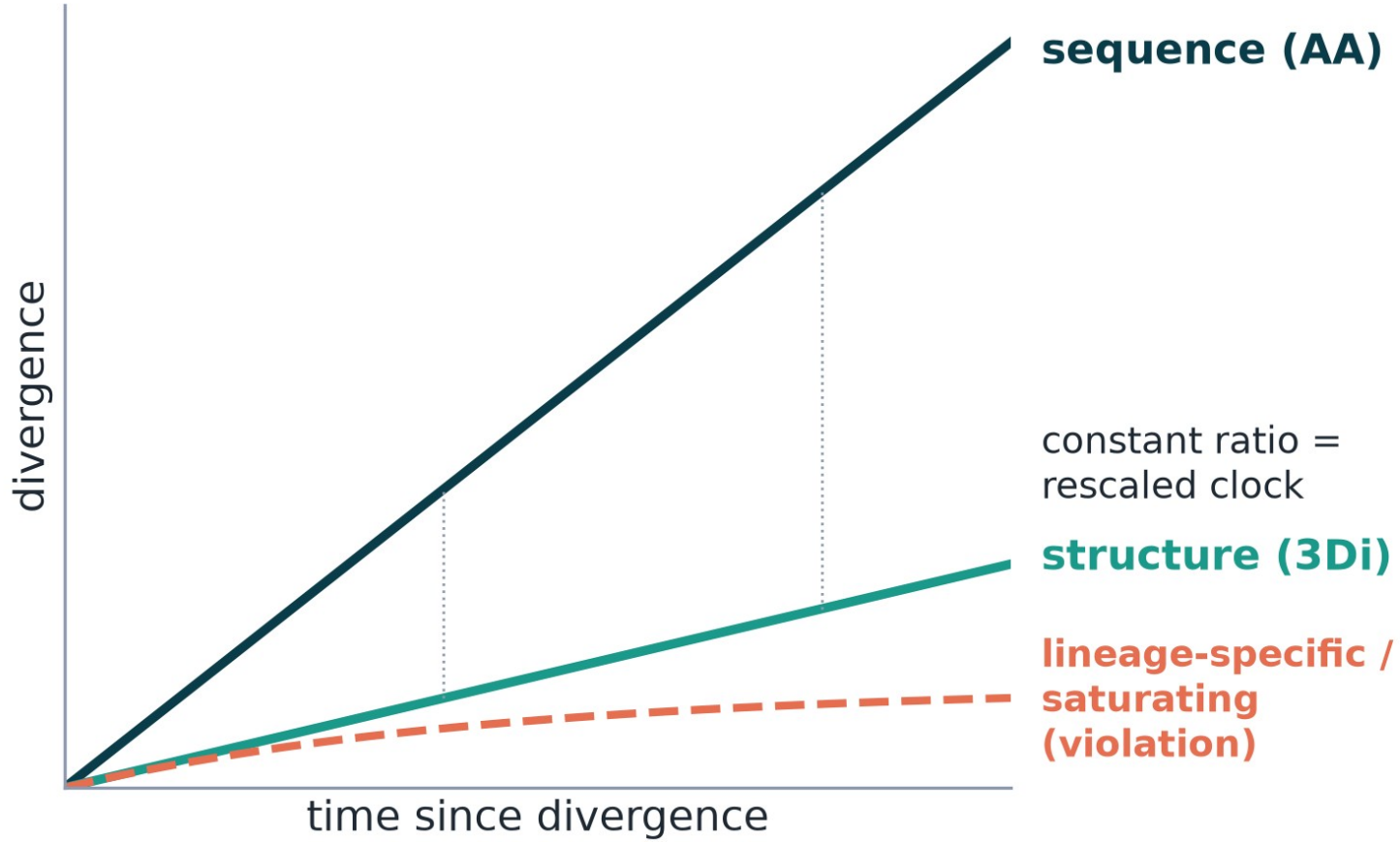
Shape

curvature c from pairs (E1, E2)

Ratio · time · lineage

branch lengths on the species tree (E3–E5)

Explainer: molecular clock and the neutral null



Proportional

constant 3Di/AA ratio across ages and lineages

Violation

ratio drifts with age, or some lineages change structure faster

How this differs from previous approaches

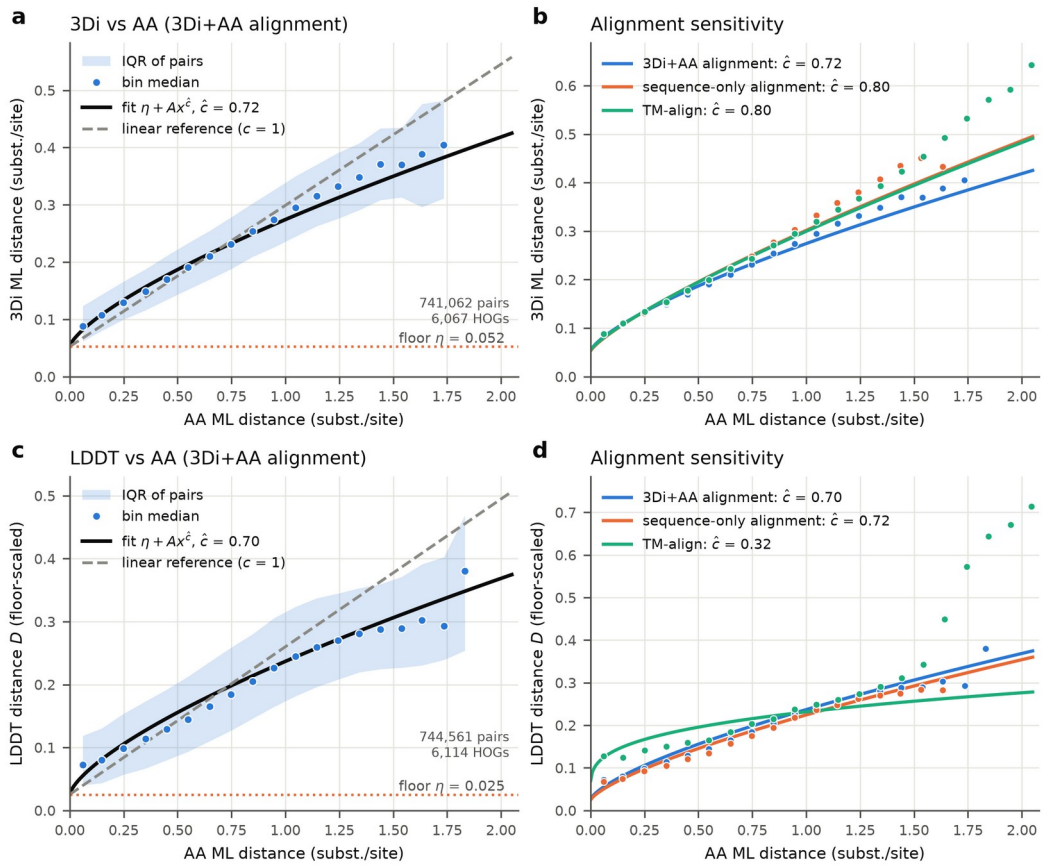
Study	Pairs	Divergence axis	Structure measure	Scale / focus
Illergård et al. 2009	SCOP domain pairs (PDB)	substitutions	core states / RMSD	~hundreds of families
Pascual-García et al. 2019	superfamilies (PDB)	substitutions	structural distances	clock violations vs function
Barrio-Hernandez et al. 2023	AFDB, 214 M models	none (cluster LCA)	Foldseek clustering	no rate analysis
Moi et al. 2025 (FoldTree)	families (AFDB)	tree topology	3Di + AA distances	phylogenies
Warwick Vesztrocy et al. 2025	paralogs (AFDB)	branch lengths	Foldseek LDDT	least-diverged ortholog
This project	**orthologs, 180 eukaryotes**	**TimeTree ages + ML**	**3Di (Q.3Di.AF), LDDT, TM**	**7,000 families, pre-registered**

5

The hypotheses, one by one

Each with: what it is, prior literature, pros and cons of our test, and the verdict

H1 · Shape: does structure diverge linearly with sequence?



3Di (and LDDT) distance vs amino-acid distance across ortholog pairs; curvature c = 1 is linear.

Prior work

Chothia & Lesk 1986 (exponential vs identity); Koehl & Levitt 2002; Illergård et al. 2009 (linear vs ML distance).

+ Pros

- orthology-selected pairs
- matched 20-state models
- errors-in-variables correction
- 3 alignments

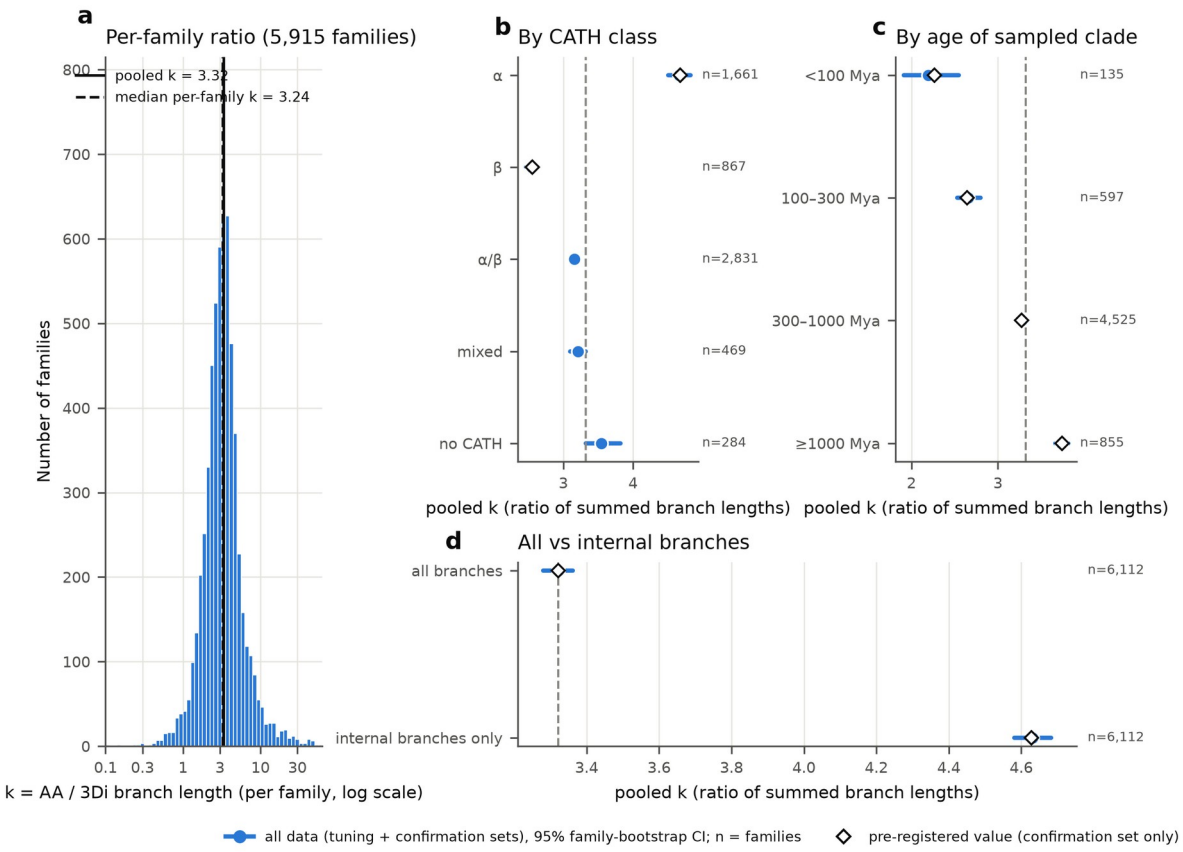
– Cons

- per-model AlphaFold noise biases toward concave
- mild curvature hard to detect

Pre-registered — Concave: ĉ = 0.72 [0.64–0.80], robust to alignment

Post hoc (verified) — **Curvature bracketed 0.75–1.0; moves toward 1 in high-confidence models**

H2 · Is structure 3–10× more conserved than sequence?



k = total amino-acid change / total 3Di change over tree branches.

Prior work

Illergård et al. 2009: 3–10× after alphabet matching (15–30× with raw states); β faster than α.

+ Pros

- true 20 vs 20 comparison
- branch-level sums on the species tree
- 4,891 families

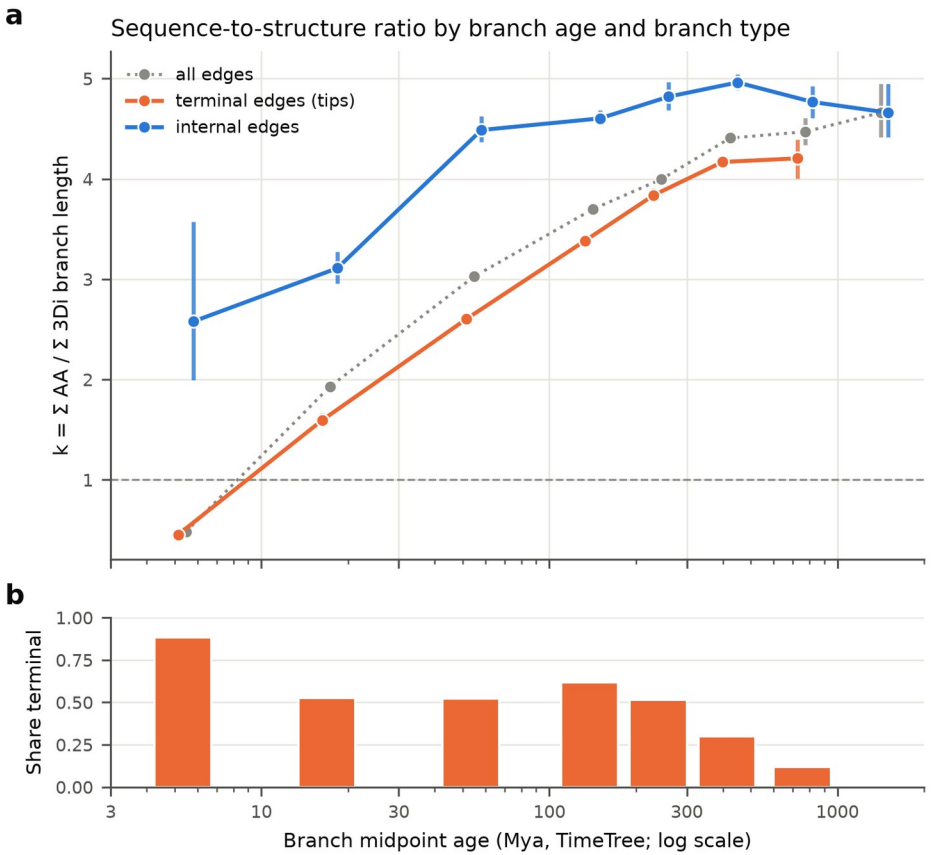
– Cons

- specific to 3Di and predicted structures
- terminal branches inflated by noise

Pre-registered — k = 3.3 [3.09–3.57]; α 4.7× vs β 2.6×

Post hoc (verified) — **Internal branches: k = 4.6; α 6.1× vs β 3.9× (gap persists)**

H3 · Time: is the structure/sequence ratio constant?



Ratio k by branch age (TimeTree), split into terminal (species tips) and internal branches.

Prior work

Zuckerkandl & Pauling 1965; Kimura 1968 (neutral clock); Pascual-García 2019 vs Moi 2025 (mixed evidence).

+ Pros

- dated species tree
- alternative deep timescales
- terminal vs internal split

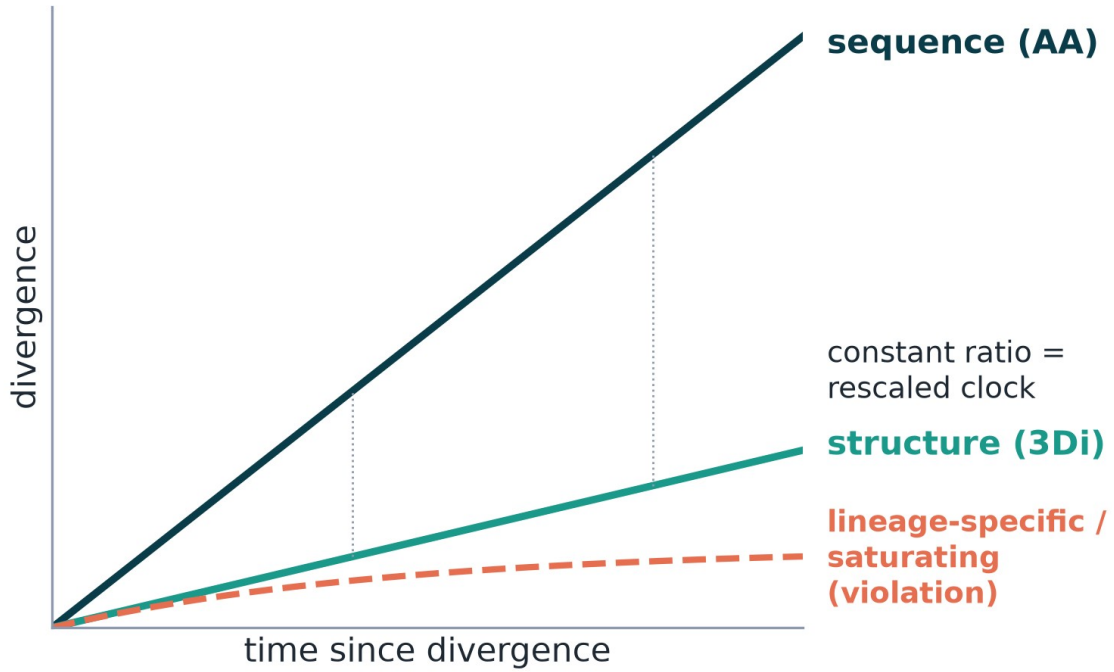
− Cons

- only 224 young families
- deep dates uncertain
- tip-correction sensitive

Pre-registered — Ratio rises with age (p_Holm 3×10^{-12}), robust to α and timescale

Post hoc (verified) — **Age trend = terminal-branch noise artefact; internal ratio flat ≈4.6 from 30 My to >1 Gy**

H4 · Lineages: is there a neutral structural clock?



Do some lineages change structure faster than their sequence change predicts?

Prior work

Ohta 1973 (nearly neutral); Bromham & Penny 2003 (rate variation); Pascual-García 2019 (structural clock violations).

+ Pros

- sister-lineage contrasts
- spike-in power test
- guards against artefacts

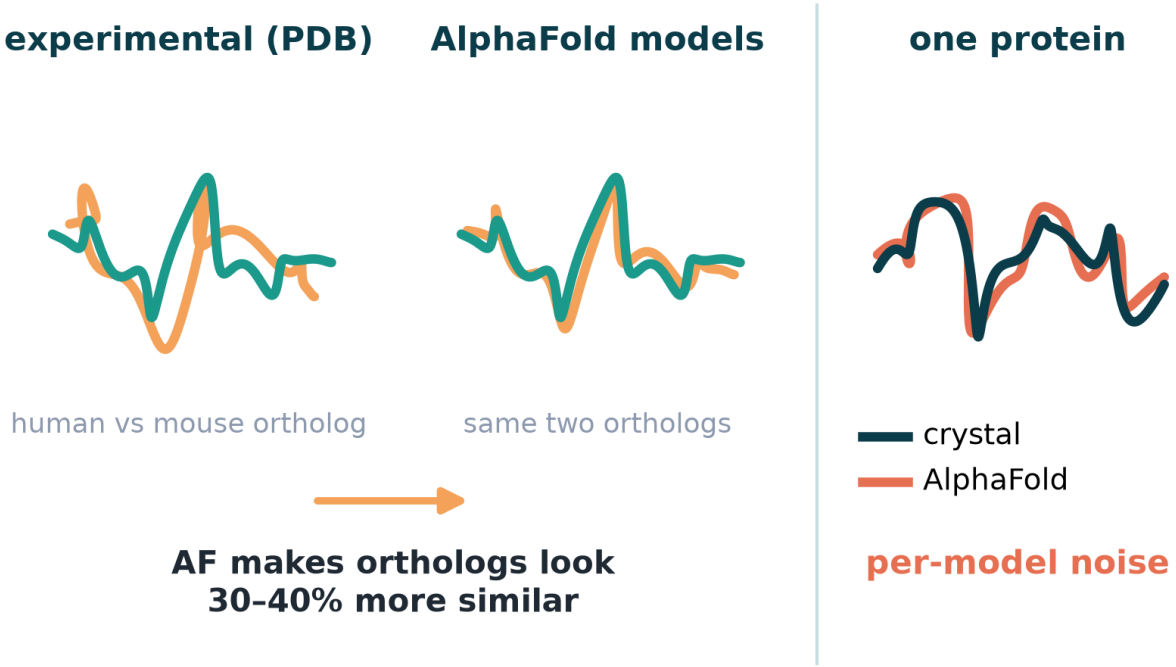
− Cons

- power only 24%
- accepting the null practically unreachable

Pre-registered Artefact-suspect: excess vanishes on internal branches

Post hoc (verified) **No robust lineage-specific structural rates; neutrality not testable with this design**

H5 · Do AlphaFold models make orthologs look too similar?



Prior work

Jumper et al. 2021; Buel & Walters 2022 and Pak et al. 2023 (AF insensitive to point mutations).

+ Pros

- identical residue sets
- post-training-cutoff structures
- 197 post-cutoff pairs

– Cons

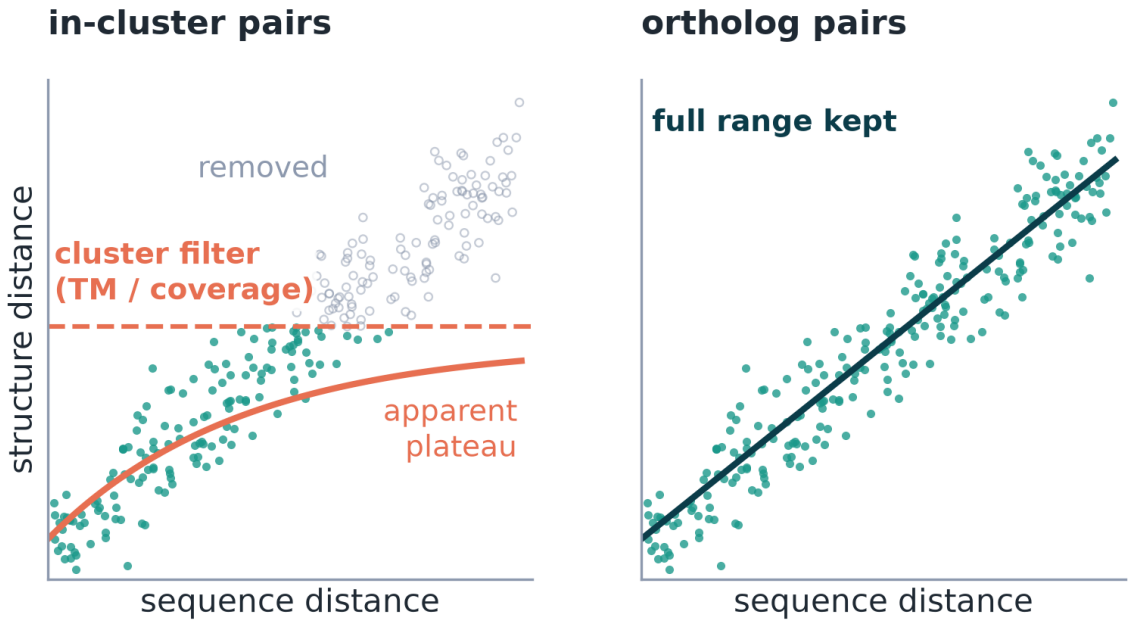
- few deep pairs
- crystal conditions add own noise

Exploratory

AF–AF divergence = 0.59× PDB–PDB for 3Di

Same ortholog pairs and same residues: AlphaFold vs experimental (PDB) structures.

H6 · Do structural clusters hide divergent pairs?



Prior work

Barrio-Hernandez et al. 2023 (clustering: 90% coverage, E < 0.01).

+ Pros

- 27.7 M cluster pairs vs our orthologs
- same metrics

– Cons

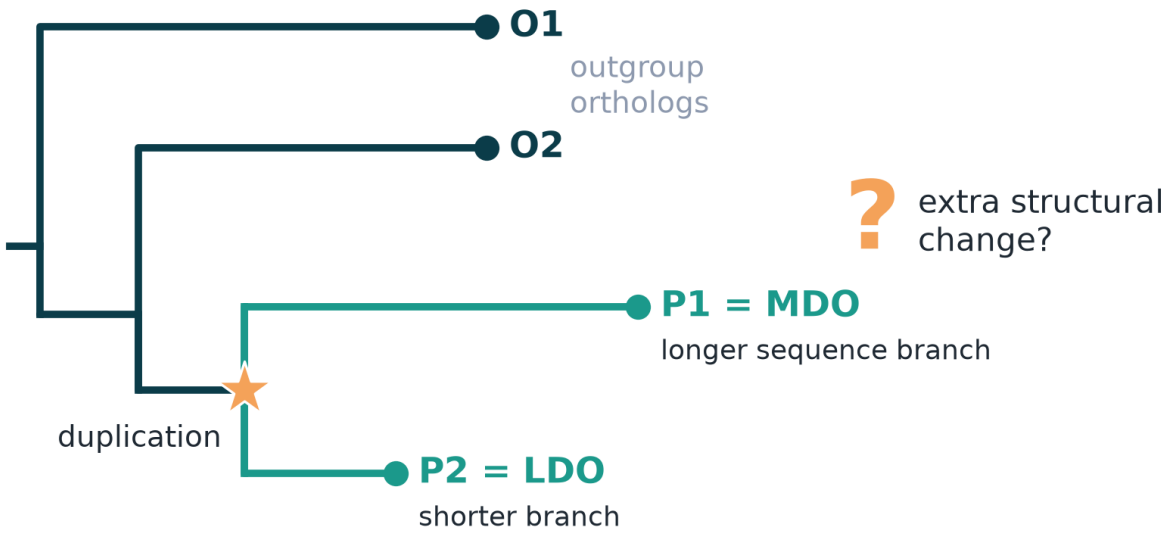
- ranges barely overlap (10–35% vs 47–99% identity)
- scoring differs

Exploratory

in-cluster TM plateaus ≈0.83 = prediction-quality ceiling

Pairs inside AFDB clusters passed a structural filter; ortholog pairs did not.

H7 · Duplication: do paralogs diverge more in structure?



Positive control and least-diverged-ortholog (LDO) test with split alignment columns.

Prior work

Ortholog conjecture: Nehrt et al. 2011, Altenhoff et al. 2012; LDO: Warwick Vesztrocy et al. 2025.

+ Pros

- design avoids regression to the mean
- 3,525 families

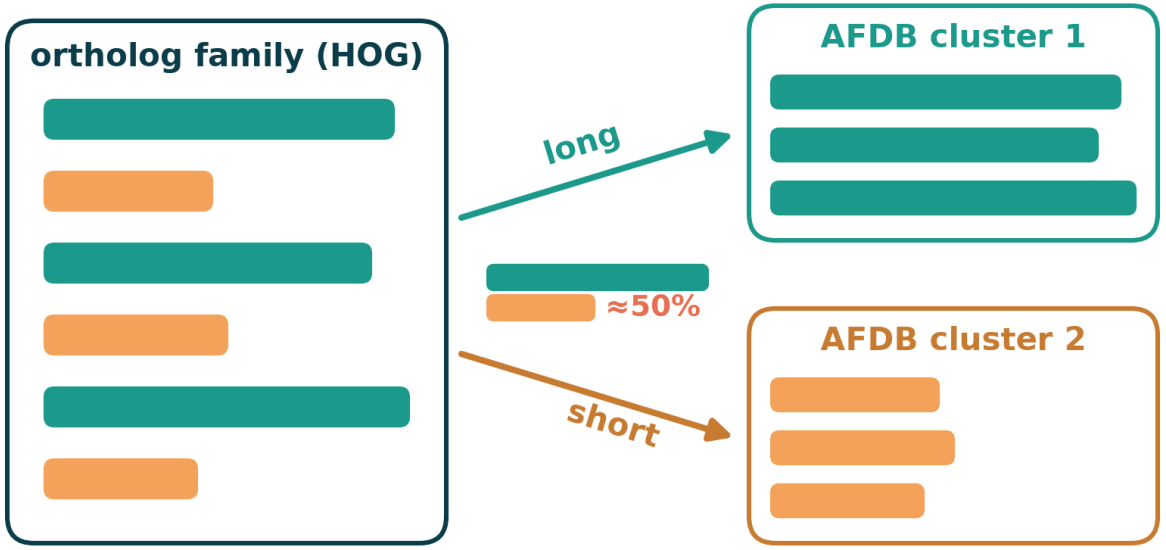
− Cons

- paralog part not alignment-robust
- effect ~ size of per-model noise

Pre-registered Detected: paralog excess 0.030; faster copy +0.017

Post hoc (verified) **Faster-copy excess halves after pLDDT adjustment (0.011) but stays positive; direction replicates**

H8 · Are AFDB structural clusters ortholog groups?



✗ Foldseek clustering needs ≥ 90% coverage of both proteins

Prior work

Barrio-Hernandez et al. 2023 (cluster ages by last common ancestor); TED domains (Lau et al. 2024).

+ Pros

- all families
- logistic model of splits

– Cons

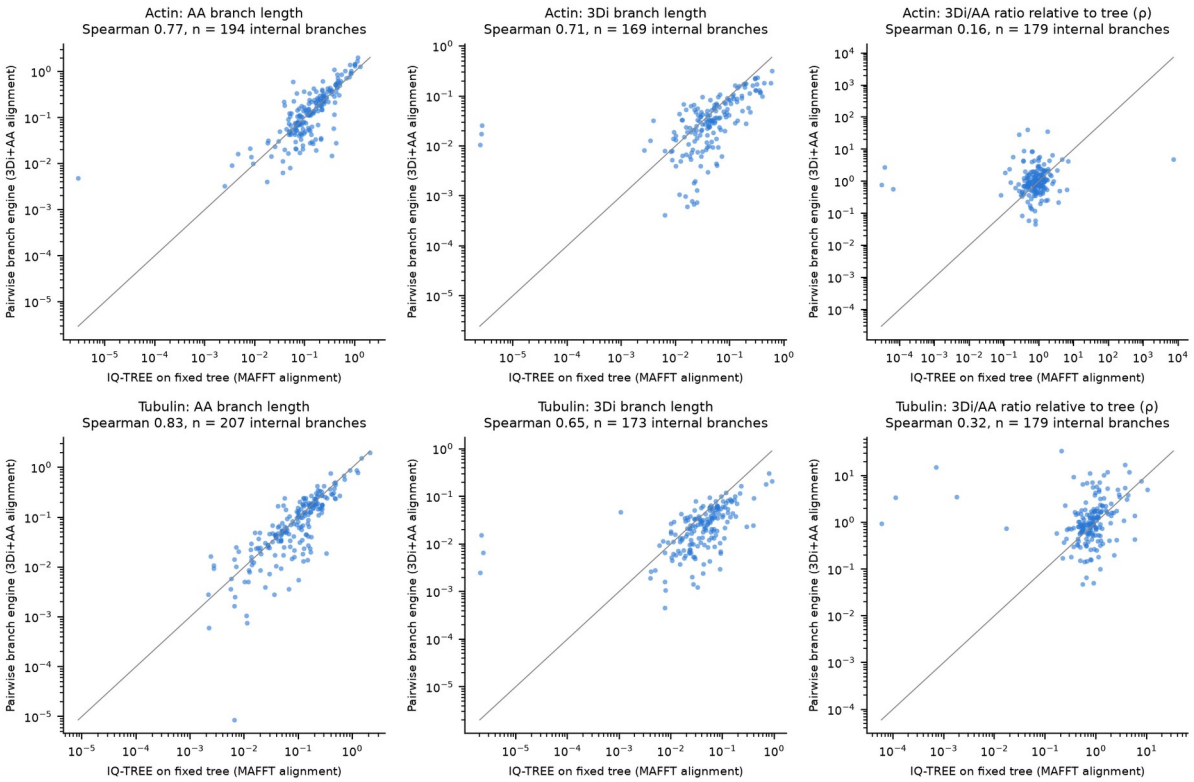
- Ancestra orthology itself unverified
- domain labels noisy (21–25%)

Exploratory

70% of groups split across clusters

Map 109,598 ortholog groups onto AFDB structural clusters.

H9 · Case studies: bursts of structural change?



Actin/tubulin folds (with prokaryotic & Asgard homologs) and protein kinases.

Prior work

Prokaryotic cytoskeleton homologs (MreB, FtsZ), Asgard archaea; kinase evolution (Kannan & Taylor).

+ Pros

- fixed reference trees
- 3 alignments
- two estimators

– Cons

- per-branch ratios disagree between methods
- deep sequence saturation

Exploratory

no robust stem bursts

6

Results and take-home messages

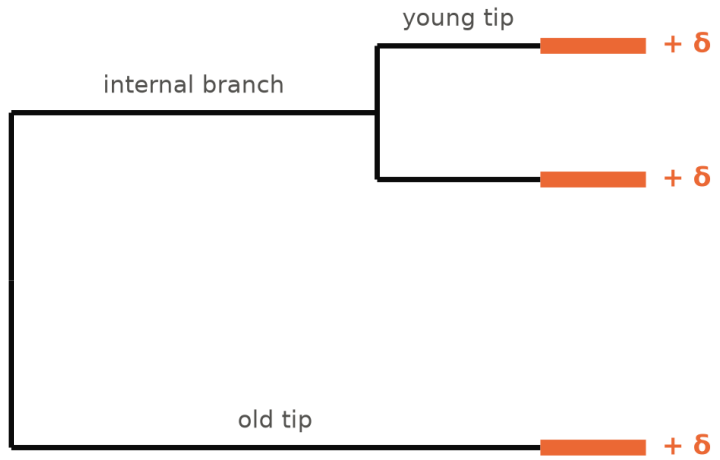
What the pre-registered tests said, and what survived scrutiny

Scorecard: pre-registered vs post-hoc reading

Question	Pre-registered (confirm set)	Best reading
Shape (E1/E2)	concave, ĉ 0.72	**0.75–1.0: near-proportional**
Ratio (E3)	3.3× (α 4.7, β 2.6)	**4.6× on internal branches**
Time (E4)	ratio rises with age	**flat 30 My – >1 Gy (tip artefact)**
Lineages (E5)	artefact-suspect	**no robust lineage effects**
Duplication (E6)	detected	**direction robust; size ~halved**
AlphaFold bias (X4)	—	**orthologs look 30–40% more similar**
Clusters vs orthology (X5)	—	**70% of groups split, by length**

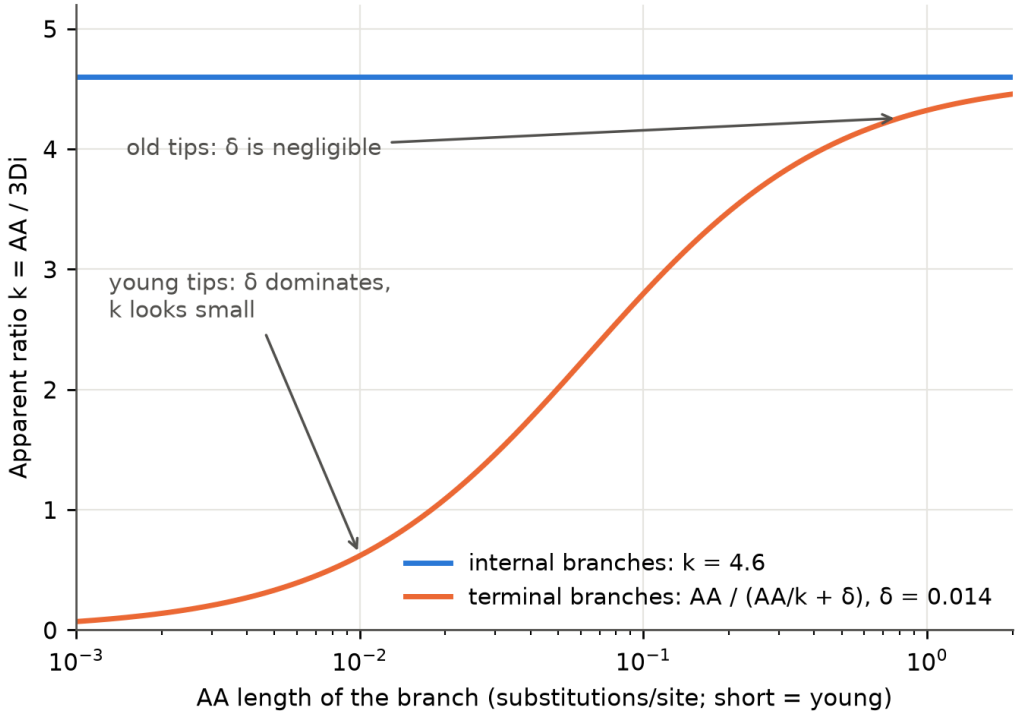
The twist: one hidden noise source explains most signals

a Every terminal branch carries the same extra 3Di length



Black: real change; 3Di length = AA length / k.
Orange: the same extra 3Di length δ on every tip (one AlphaFold model per tip, beyond the fixed correction), whatever the tip's age.

b So k appears to rise with branch age



Most young branches are terminal, so an age trend fitted over all branches turns negative (3Di/AA falls with age) even if the true ratio is constant.

Bends E1 toward concave

Creates E4 age trend

Fakes E5 lineage effects

What is new, and how it relates to the literature

Confirmed

- Structure changes more slowly than sequence (Illergård 2009): 3–5×
- β-rich folds change structure faster than α (Illergård 2009)
- Direction of the LDO structural result (Warwick Vesztrocy 2025)

Refined

- 'Linear' holds roughly, now on matched 20-state models and thousands of families
- Structure is proportional to sequence on a real time axis (30 My – >1 Gy, internal branches)
- Structural clock violations not robust once artefacts handled

New

- AlphaFold homogenises orthologs (30–40% more similar than crystals)
- Per-model 3Di noise inflates terminal branches
- AFDB clusters split ortholog groups, mostly by length

Take-home messages

≈4.6×

sequence changes faster than 3Di structure (internal branches)

0.75–1.0

curvature: structure roughly proportional to sequence, mild saturation at most

30–40%

AlphaFold makes orthologs look more alike than experiments

70%

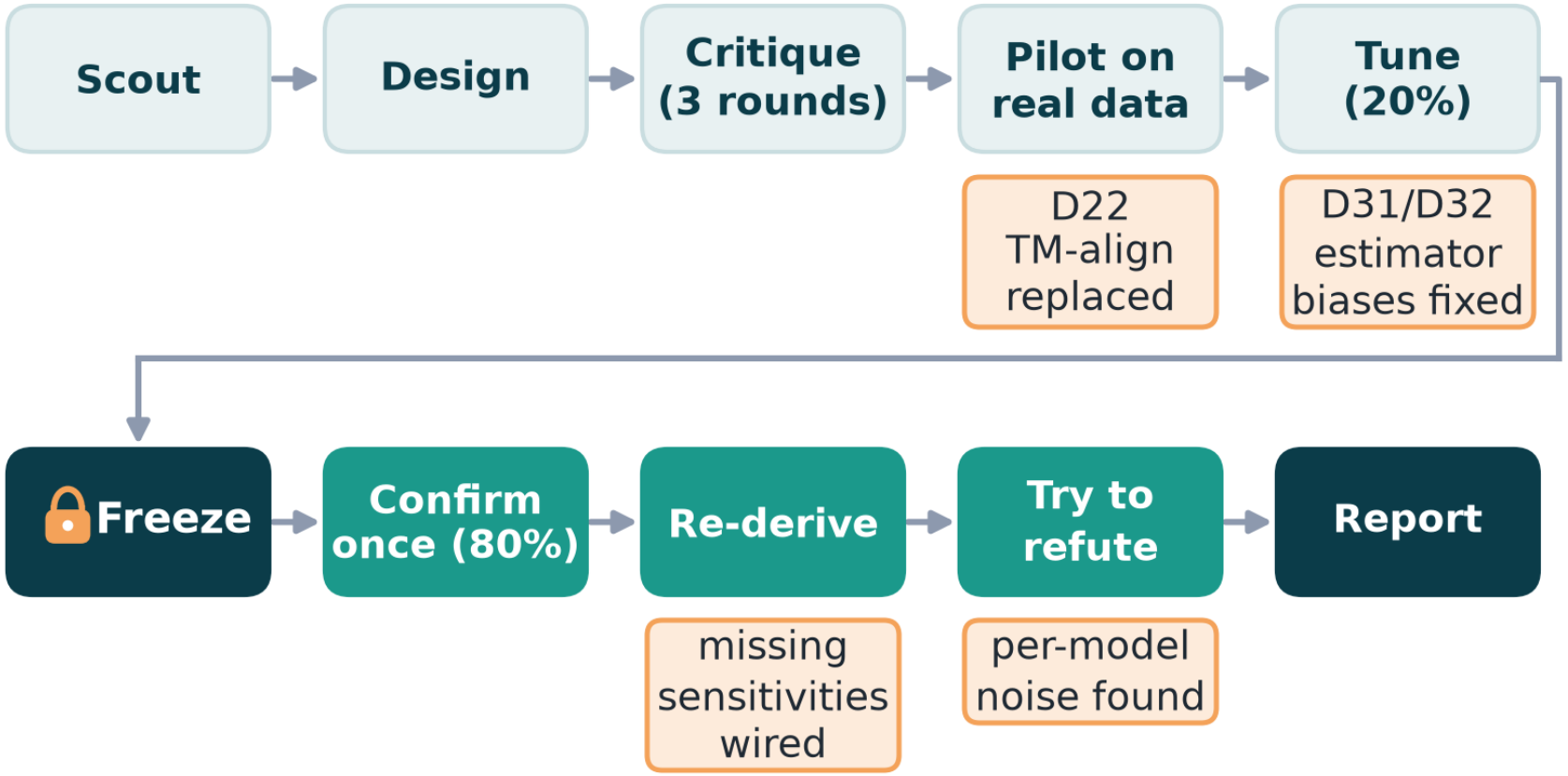
of ortholog groups are split across AFDB structural clusters

7

Decision making and the future

What transfers to the next AI-led project

The decision loop, and where it changed course



What worked, and what did not

Worked

- Parallel scouting before design
- Small real pilot before freezing
- Synthetic worlds with known answers
- Pre-registration + deviation log
- Independent re-derivation and refutation attempts
- Research tree as memory across a node move

Did not

- Noise floor from a single control type
- Plan-specified sensitivities unwired until the dry run
- Exploratory work competing with the critical path
- Reviewers shared blind spots (same model family)
- Harness quirks: blocked file writes, self-matching wait loops

Carrying the decision process to future projects

1. Scout data, tools and literature in parallel before designing
2. Pilot the whole pipeline on small real data before freezing
3. Test every estimator on synthetic truths, incl. nuisance noise
4. Validate noise floors with two independent controls
5. Freeze, confirm once, log every deviation
6. Ask a fresh agent to refute surprising conclusions

Conclusion: AI-led research

In ~23 compute hours an AI team designed, pre-registered, ran and verified a study of 1.4 M protein pairs, and corrected its own headline.

Limits: shared blind spots across agents; no human verification yet; predicted structures only.

Next: human audit · AlphaFold runs with varied MSA depth · PDB same-protein floors · time-calibrated structural phylogenetics · agentic labs at scale

Acknowledgements and open slots

To fill in

- Hackathon dates and participants: [__]
- Human contributors / project lead: [__]
- Project repository link: [__]
- Verification status by humans: [pending]

Resources used

- AlphaFold DB & AFDB clusters (Steinegger, Beltrao labs; CC-BY 4.0)
- Foldseek, FoldMason, Q.3Di.AF
- Ancestra (hackathon Phase 1)
- TimeTree (data not redistributed)
- UniProt, PDB/SIFTS, TED, Pfam